

CADEM0019 Artificial Intelligence Applications in Structures and Materials Coursework

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1 Exploratory Data Analysis

The EDA aimed to verify dataset integrity, identify non-physical simulations, and prepare the data for surrogate modelling. All 2001 samples were confirmed to be 13×13 matrices, with a dataset shape inspection of the entries validating the read-in procedure. The global thickness distribution is approximately symmetric around 0.1 m, whereas stress exhibits a positively skewed tail approaching the 250 MPa yield limit, consistent with boundary-layer concentrations near the clamped edges (see Figure [1a]).

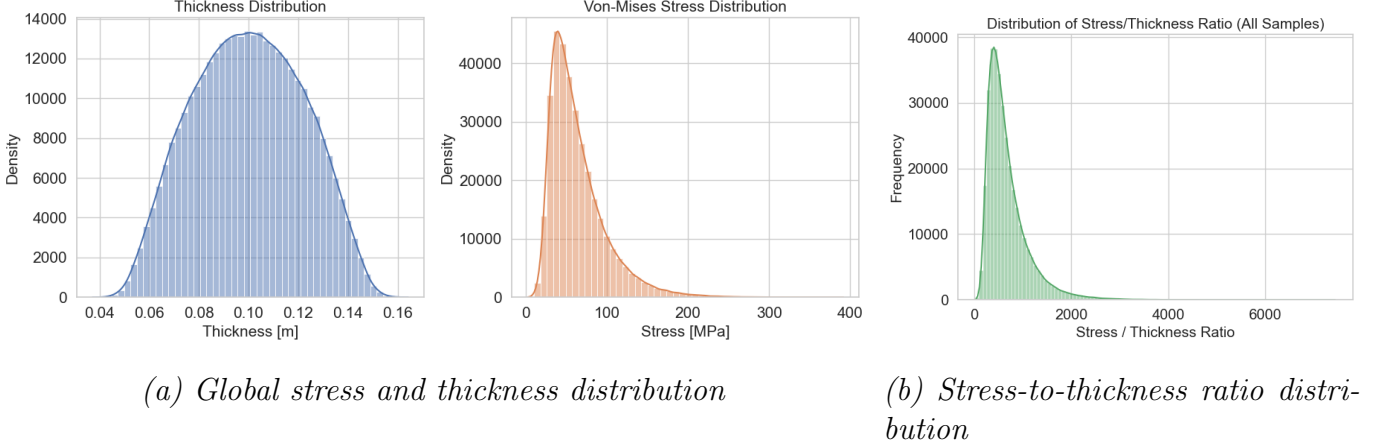


Figure 1: *Exploratory Data Analysis: global distributions and stress-to-thickness ratio.*

Because stresses exceeding yield typically indicate unstable simulations, outlier detection was performed using the *stress-to-thickness* ratio, a physically meaningful indicator due to stress’s inverse dependence on bending stiffness. The ratio histogram (Figure [1b]) revealed a strongly skewed distribution, making a classical 3σ rule unsuitable as it assumes Gaussian symmetry and risks discarding physically valid high-stress cases. A hybrid approach was therefore adopted, combining z -score screening, percentile filtering, and spatial smoothing applied to the ratio field. The 3σ rule highlighted local cell-level anomalies, while the percentile filter controlled global skew. The initial 5–95% range removed too many samples ($\approx 75\%$), so thresholds were relaxed to 0.5–99.5%. Spatial smoothing eliminated isolated discontinuities inconsistent with structural behaviour.

Violin plots (see Figure [2]) confirmed that the cleaned dataset retained consistent ratio distributions while preserving natural variability across thickness patterns. Extreme ratio values (~ 6000) were not automatically removed when confined to isolated hotspots, since these are mechanically plausible near boundaries and were verified via heatmaps. Approximately 11% of samples (221 outliers) were ultimately excluded. Finally, each sample is represented as a 13×13 spatial field for both thickness and stress, forming a consistent grid structure suitable for convolution-based surrogate modelling.

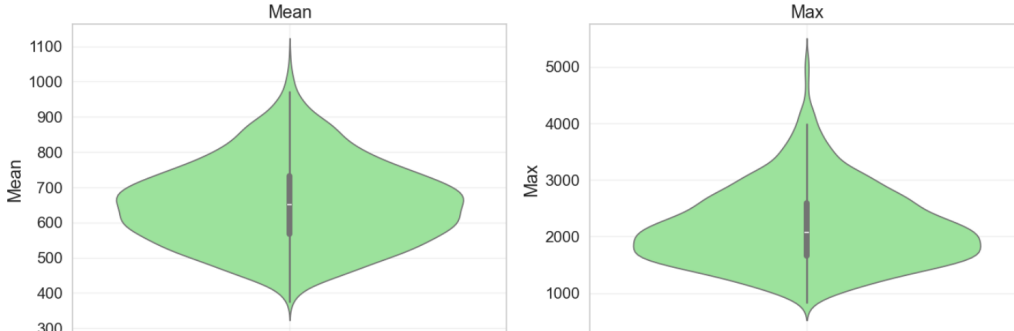


Figure 2: *Violin Plots of stress to thinkess ratio used in outlier detection*

A *Pearson* correlation check showed that mean thickness has a weak negative correlation with peak stress (-0.18), confirming its physical relevance while motivating the use of the full 13×13 spatial field for surrogate modelling.

2 Surrogate Modelling

2A. Modelling Approach

The problem requires mapping a 2D thickness distribution to a 2D *Von-Mises* stress field. Simple analytical methods like polynomial fitting provide a baseline to establish global trends. However, these methods are insufficient for capturing the spatial features required to predict the 2D stress field resulting from a plate with variable thickness. A *Convolutional Neural Network (CNN)* was therefore selected, as *CNNs* are designed to capture spatial correlations through local filters, making them well-suited for full-field stress prediction. Stress patterns depend strongly on local thickness variations, analogous to how convolutional kernels operate on neighbouring regions. The model comprises *Conv2D* layers, followed by *Flatten*, *Dense*, and a final *Reshape* layer to output a 13×13 stress field.

The model produces a feature hierarchy, allowing an increasingly complex learning as the data is processed layer by layer. *CNNs* do not require data flattening, making them less susceptible to overfitting than dense networks when handling spatial data. *Autoencoders (AEs)* are a plausible alternative, as they learn compressed spatial representations. However, their objective is to minimise reconstruction error rather than optimise pointwise stress prediction. The task prioritises accurate stress prediction rather than latent-space reconstruction. Hence, the *CNN* approach was retained as a more direct and intuitive solution for full-field stress prediction. This avoids the additional complexity of encoding and decoding a latent space while efficiently handling the spatial data.

2.B Training & Validation Results

The model used a standard *Train(70%)/Validation(15%)/Test(15%)* split to evaluate the *CNN* surrogate model; using 1246 training samples. Random splitting was retained to ensure representative coverage of low and high-stress samples in validation and test partitions. The validation set monitors the *early stop* functions and is used to prevent overfitting of the model. Hence, a 15% split for this batch was retained necessary. The model used an *Adam* optimiser with a learning rate of 1×10^{-3} . The model was programmed to run a total of 200 epochs; however, the *early stop* function interrupted the training at 119 Epochs, after a learning rate plateau of 6.25×10^{-5} was reached (see Figure [3]).

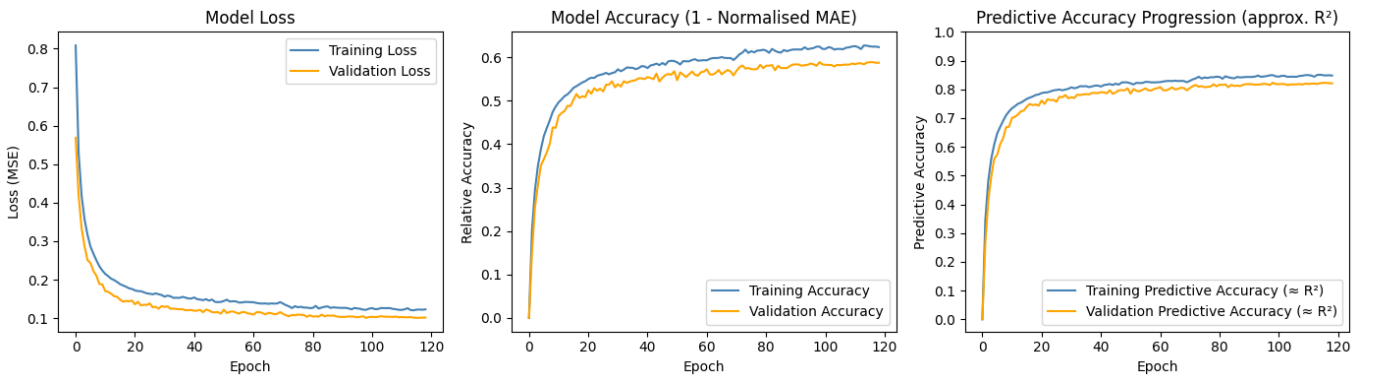


Figure 3: *Training history of the Surrogate model monitoring the progression of loss (MSE), MAE, and predictive accuracy (approx. R^2) across epochs*

The training process uses the validation set to monitor performance using validation loss metrics, adjusting the learning rate when the loss plateaus and stopping training early if the loss does not

improve after 20 epochs. The first graph shows an exponential and nearly identical decrease in training and validation MSE , confirming effective learning without overfitting. The absence of divergence indicates stable optimisation dynamics and confirms that the model generalises well rather than memorising the training data. The central graph shows accuracy increased rapidly in the first 20 epochs, then gradually plateaued, showing a good generalisation is maintained (training-validation gap < 0.1). The final graph shows a training-time proxy for predictive accuracy (computed as 1 minus the normalised MAE), gradually converging towards 0.84. This value is not the formal statistical R^2 but provides a useful visual indication of improving predictive capability during training. The true R^2 , evaluated on the independent test set, is reported separately in Table [??]. The exponential learning-rate decay indicates stable convergence, and the close alignment between training and validation curves confirms consistent feature extraction across both datasets. After training was complete, the model’s performance was validated against the unseen *Test* set. The performance metrics using transformed stress values were reported in Table [1]:

Table 1: *Performance Metrics on Test Dataset*

Metric	Result
Mean Squared Error (MSE)	81.77 MPa
Root MSE	9.04 MPa²
Mean Absolute Error (MAE)	6.23 MPa
Coefficient of Determination (R^2)	0.928

Furthermore, a visual comparison of the model’s prediction was made, providing further verification (see Figure [4]). A highly accurate model prediction of stress patterns and values can be seen.

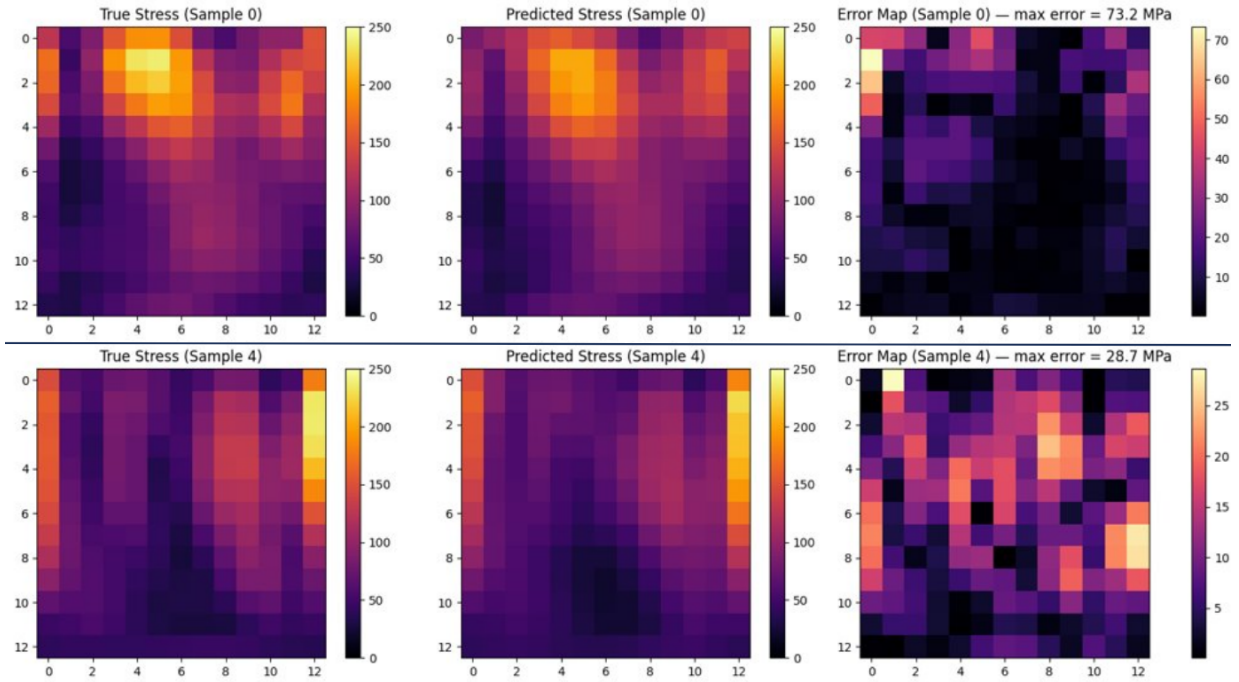


Figure 4: *Comparison between predicted and true stress samples*

An MAE of 6.23 MPa (= 7% of yield) represents only 3–4% of the upper stress range. Furthermore, the hotspots confirmed a slight underprediction in stress. Given the design constraint of no-yield ($s < 250$ MPa), even modest underprediction near hotspots can misclassify unsafe designs as feasible. As a final performance index, achieving a final predictive accuracy of 92.3% suggests the *CNN* captured a large proportion of variability between thickness distribution and the *Von-Mises* distribution. Therefore, aligning with the robust approach carried out. Some residual variance remains unexplained, likely due to sharp gradients in proximity of fixed boundaries. The MSE tends to penalise large local

errors providing further insight on the model’s performance. A *Root Mean Squared Error (RMSE)* of 9.04 MPa is acceptable considering typical deviations are within 10 MPa (1). Hence, meaning no catastrophic mispredictions (of $30 - 50 \text{ MPa}$) will occur. Regardless, subsequent robustness checks were performed to address the previously discussed underestimation of stresses.

2.C Model Robustness Investigation

All samples whose maximum *Von-Mises* stress exceeded 85% of the material’s Yield stress (212.5 MPa) were subject to a robustness evaluation. For these cases, the surrogate must accurately capture both the location and magnitude of peak stresses. Despite achieving high accuracy, one can see the model has a consistent negative bias, particularly in regions of high stress concentration (see Figure [5]). This is a typical trait of *Neural Networks* where smooth outputs are favored, suppressing sharp peaks. This trait can be explained by the training loss *MSE* and *MAE* penalising large errors rather than small ones, pushing the network to more conservative solutions. While the general bias magnitude remains modest (-10.66 MPa), underprediction of yield-critical behaviour poses a structural-safety concern: it may lead to false feasibility classification during optimisation. This behaviour contrasts with typical engineering safety philosophy, which favours conservative overestimation rather than underprediction. The peak stress bias was later numerically evaluated by plotting trends in errors of high stress cases (Figure [5]).

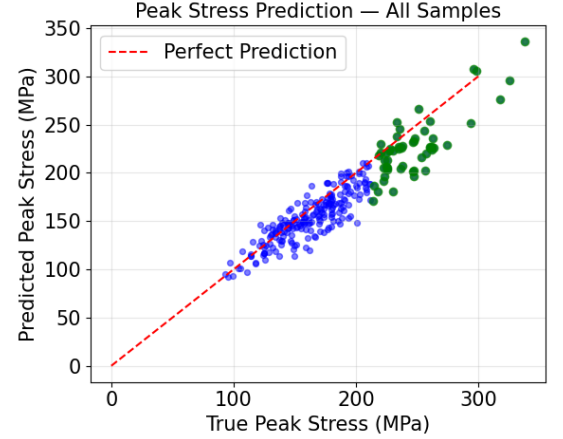


Figure 5: *Bias investigation through peak stress prediction (green).*

A large proportion of data points can be seen to lie below the line traced by an ideal model. Resulting in: an average peak stress bias of -16.92 MPa , confirming the visual and numerical evidence provided prior to the robustness check. This confirms the model’s systematic tendency to underpredict yield-critical stresses. Underprediction may cause feasible/infeasible misclassification in optimisation, particularly near the 250 MPa constraint. Ensuring accurate peak prediction is essential to avoid unsafe or suboptimal results. To overcome this defect the application of a weighted loss function should be considered(2).

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N w_i |\sigma_i - \hat{\sigma}_i|, \quad w_i = 1 + \alpha \frac{\sigma_i}{\sigma_{\max}}. \quad (1)$$

This adjustment directly supports the optimisation task, as it reduces the risk of yielding by improving surrogate accuracy in high-stress regions. By increasing the contribution of high stress cells to the computed loss, the model’s training will be enhanced. Such approaches are well-established in deep learning (3) and directly address optimisation requirements by improving peak-stress fidelity. The bias was mitigated by adding the bias magnitude to the predicted stresses with the addition of MAE, providing a simple yet effective estimate.

3.1 Beam Optimisation

3.1.A-B Problem Formulation

The optimisation aims to minimise the mass of a rectangular hollow beam of fixed length $L = 1 \text{ m}$ under stress, displacement, and geometric constraints. The beam’s optimised parameters were the following: outer width b , outer height h , and wall thickness t illustrated in Figure [6].

Mass is proportional to the cross-sectional area, forming the objective function:

$$f(x) = A(b, h, t) = bh - (b - 2t)(h - 2t) \quad (2)$$

Structural constraints derive from *Euler–Bernoulli* beam theory, using the second moment of area $I(b, h, t)$ to compute bending stress $\sigma_{\max}$, and tip deflection $d(b, h, t)$:

$$\sigma_{\max}(b, h, t) = \frac{PL \left(\frac{h}{2}\right)}{I(b, h, t)} \quad (3)$$

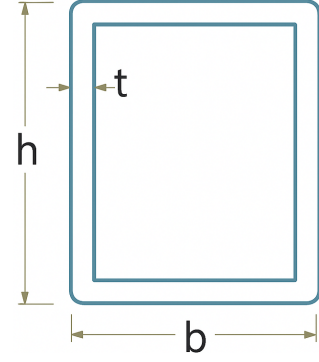


Figure 6: Diagram of beam cross section

The constraints applied for optimisation are represented in Table [2]:

Structural Constraints	Equation
Maximum bending stress	$\sigma_{\max}(b, h, t) \leq \sigma_y$
Tip deflection limit	$\delta(b, h, t) \leq \delta_{\max}$
Geometric Constraints	Equation
Wall thickness validity (height)	$h - 2t \geq 0$
Wall thickness validity (width)	$b - 2t \geq 0$
Coursework geometric rule	$h - 2b \geq 0$
Thickness ratio constraint (height)	$t \leq \frac{h}{20}$
Thickness ratio constraint (width)	$t \leq \frac{b}{20}$

Table 2: Structural and geometric constraints used in the Q3.1 beam optimisation problem.

Using a gradient-based *Trust-Constr* algorithm, the constraints were expressed in the following vector form:

$$g_1(x) = 1 - \frac{\sigma_{\max}(x)}{\sigma_y}, \quad g_2(x) = 1 - \frac{\delta(x)}{\delta_{\max}} \quad (4)$$

A constrained optimisation algorithm such as *Trust-Constr* is necessary as the *Karush-Kuhn-Tucker* (*KKT*) conditions must be satisfied at the optimum. *Trust-Constr* is well suited to tightly constrained engineering problems, preserving feasibility throughout and handling non-linear boundaries reliably. Hence, making this a robust approach, optimal for the inequality constraints the task holds. In contrast, unconstrained *ML* optimisers such as *Adam* or *SGD* cannot enforce inequality constraints and may violate geometric limits.

3.1.C-D Numerical Implementation & Gradient Analysis

Three distinct initial conditions were used to test the sensitivity to starting points on the existing domain: two lying on both extremities of b and t constraints, and the latter equidistant from both limits. Analytical gradients were not considered because the constraint vector was large and contained many non smooth components, producing non-differentiable boundaries where constraints activate. Hence, making this approach error-prone.

Gradients of both the objective and constraints (df/dx and dg/dx) were obtained using forward finite differences (gradient evaluation), matching the laboratory session’s approach. A step-size sweep (1×10^{-2} to 1×10^{-6}) was performed to assess numerical stability (see Table [3]).

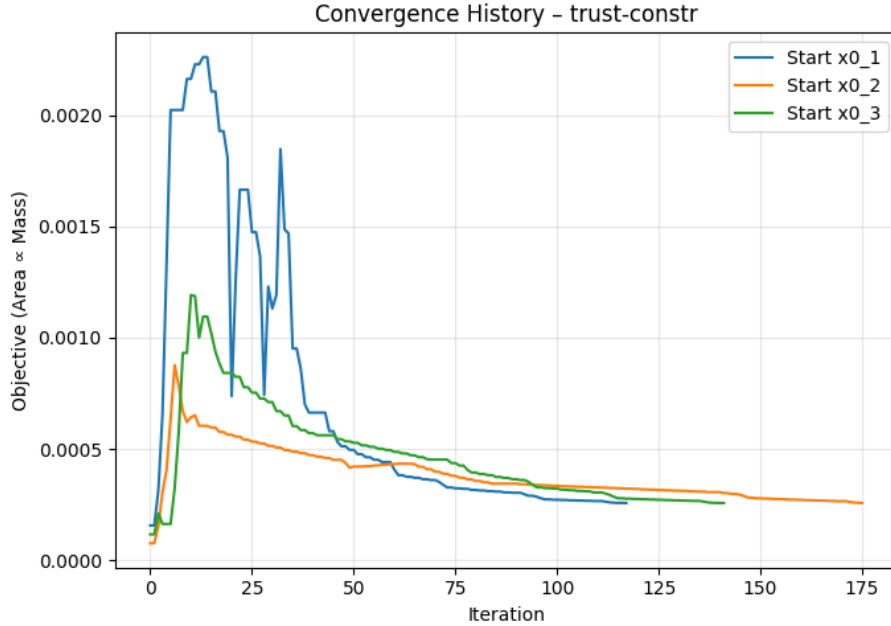
Table 3: *Constraint Gradient Sensitivity for $g_1(x)$*

Step Size h	grad_b	grad_h	grad_t	$\text{grad}_{A(t)}$
10^{-2}	17.2297	20.3862	75.6006	0.1272
10^{-3}	20.4121	23.9500	363.1713	0.1632
10^{-4}	20.7962	24.3710	571.5004	0.1667
10^{-5}	20.8354	24.4184	606.1488	0.1671
10^{-6}	20.8393	24.4181	609.8449	0.1671

Large steps (1×10^{-2} to 1×10^{-3}) introduced truncation error, while very small steps (1×10^{-6}) amplified floating-point noise. Stable and consistent gradients were obtained at $h = 1 \times 10^{-5}$, where both objective and constraint derivatives changed minimally between successive refinements. This confirmed the derivative accuracy required for stable convergence. Furthermore, indicating a consistent convergence behaviour despite being near active constraints. Thus, explaining the results obtained in the following section.

3.1.E Optimisation Results

The convergence history plot in Figure [7] shows the optimiser’s ability in effectively finding the optimum. Small variations in convergence rate were observed, with starting points closer to the geometric upper boundary (x_{o1} & x_{o2}) converging more rapidly.


 Figure 7: *Convergence plot of the three initial conditions*

The optima from the three initial conditions converged to very similar beam configurations:

 Table 4: *Initial Guesses & Optimised Dimensions*

Case	x_0 (m)	b^* (m)	h^* (m)	t^* (m)
$x_{0,1}$	[0.03, 0.05, 0.001]	0.02975	0.05976	0.001482
$x_{0,2}$	[0.01, 0.03, 0.001]	0.02974	0.05974	0.001486
$x_{0,3}$	[0.02, 0.04, 0.001]	0.02975	0.05975	0.001483

A 2D slice at a fixed value of $t = 1.5\text{mm}$ is illustrated in Figure [8a], corresponding to the optimal wall thickness obtained (to the nearest 0.1mm).

Table 5: Resulting Structural Performance

Case	Mass $\sim A$ (m ²)	$\sigma_{\max}$ (MPa)	δ (mm)
$x_{0,1}$	2.566×10^{-4}	248.7	39.63
$x_{0,2}$	2.571×10^{-4}	248.3	39.59
$x_{0,3}$	2.566×10^{-4}	248.7	39.64

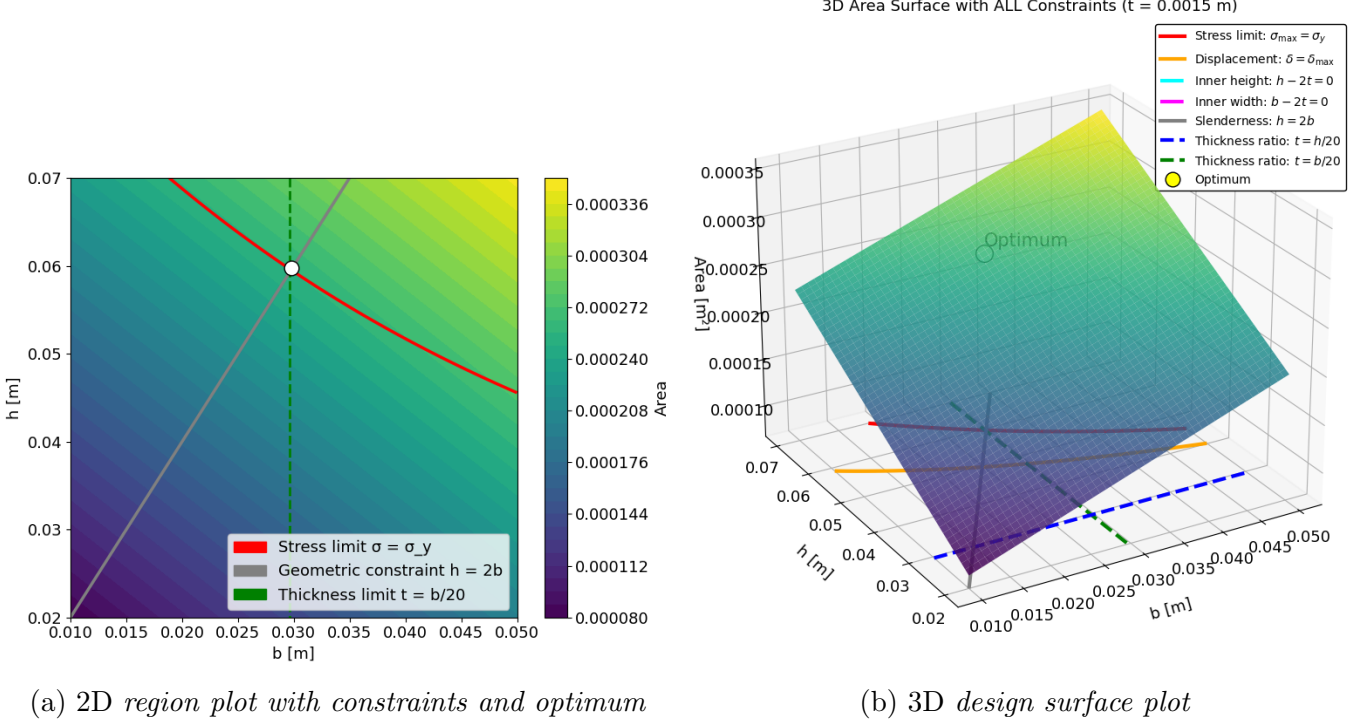


Figure 8: Constraint map (left) and objective surface (right). Optimum is shown as a white circle

3.1.F Feasible Region Interpretation

Figure[8a] shows that the feasible region in the (b, h) design space is highly restricted, primarily due to the non-linear bending-stress limit (4). As the moment of inertia decreases, the stress boundary curves upward, producing the characteristic bow-shaped upper limit. Hence, eliminating the entire low-area bottom-left region where the objective would otherwise descend. The geometric slenderness constraint $h = 2b$ forms the dominant linear boundary, creating a narrow, wedge-shaped feasible region.

Within this corridor, the optimiser attempts to reduce both b and h , but progression towards the low-area region is blocked by the curved stress limit. As a result, the optimum lies at the intersection of two active constraints: the stress limit $\sigma_{\max} = \sigma_y$ and the geometric relation $h = 2b$. Other constraints—including displacement ($\delta = 39.6$ mm $\ll$ 70 mm), wall-thickness validity, and the thickness ratio limits remain slack and do not influence the final design.

3.1. G-H Optimum Design Discussion & Customer Recommendations

The optimum configuration satisfied all constraints, obtaining an area of 2.566×10^{-4} m² and the respective b, h and t values: 0.02975m, 0.05976m, and 0.001482m. All three runs converge to nearly identical values of b, h , and t (see Table [5]), indicating a single dominant local minimum within the feasible region. The active *KKT* set consists of the stress constraint and the geometric relation $h = 2b$, which together determine the position of the optimum. The thickness limit $t = b/20$ is approached closely but remains inactive, while the displacement constraint stays substantially slack. This behaviour reflects the strongly shaped feasible region and explains the consistent convergence

paths across all initial conditions. The narrow feasible corridor visible in the 2D slice drives the optimiser toward the same active-constraint intersection regardless of the starting point.

Since the displacement constraint is highly slack and the thickness ratio constraint is near-active, the current specification may impose unnecessary mass penalties. Relaxing the displacement limit, modifying wall-thickness rules, or selecting a material with higher yield strength would widen the feasible region and enable lighter designs. These modifications can be evaluated efficiently by re-running the optimisation with updated limits, allowing the customer to obtain a design that better balances weight and structural performance.

3.2 Surrogate Model Optimisation

3.2.A Problem Formulation

The following problem evaluates four thickness parametrisations of increasing dimensionality (1, 4, 25, 169 *DOFs*). A conservative correction was applied to the surrogate model’s prediction of previously calculated $|bias| + MAE = 37,1 \text{ MPa}$, resulting in a simple, yet realistic correction.

All cases minimised the mean plate thickness subject to a corrected stress $(\sigma_{CNN} + 37.1 \text{ MPa}) < 250 \text{ MPa}$. Thus, accounting for known surrogate underprediction, ensuring constraint feasibility. The trained surrogate is used as the stress evaluator. Such formulation was retained adequate because mass is directly proportional to mean thickness, and the stress constraint is the main design driver. The thickness range for all optimisations was chosen to match the *CNN*’s coverage to further maximise predictive accuracy ($t_{min} = 0.04129m$, $t_{max} = 0.16423m$).

3.2.B Methodology

For cases *A* to *C*, a *Sequential Least Squares Programming* algorithm was used because the problems were: low dimensional, smooth, and required the satisfaction of design constraints. For case *D*, an *Adam* optimiser was implemented to handle the high-dimensional (169 *DOF*) setting. Hence, allowing an efficient gradient-based optimisation with the *CNN*’s stress field. Furthermore, the *Adam* optimiser provided significantly faster convergence than *SGD*, making code troubleshooting more time efficient. Both methods are appropriate given their ability in handling the structure of each formulation.

For the 1 *DOF* problem, one scalar thickness was applied across all 13x13 elements. Problem *B* (4 *DOF*) was approached using a bilinear field defined by 4 corner thickness values. Design thickness variables on the 5x5 grid were smoothed to a 13x13 via spline interpolation for problem *D*. For the final full-field 169 *DOF* problem, the independent thickness at each pixel was optimised.

3.2.C Results & Interpretation

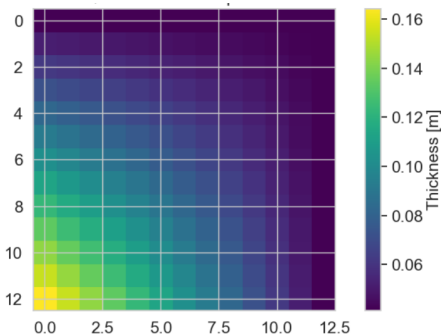


Figure 9: *Corner interpolation thickness field*

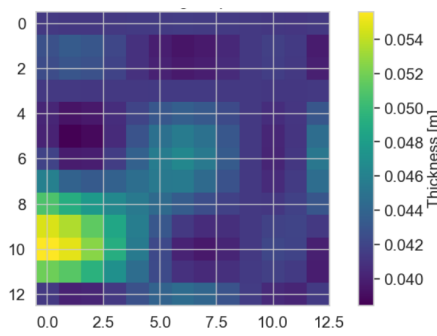


Figure 10: *5x5 grid interpolated optimum*

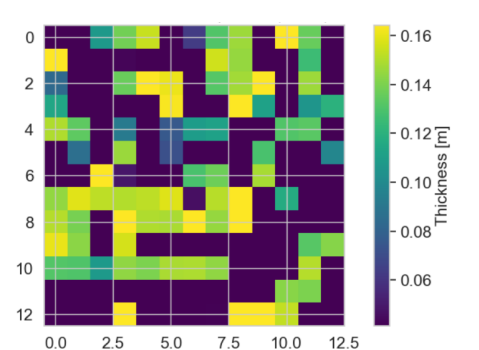


Figure 11: *Full 13x13 optimum*

Case	Mean Thickness (mm)	$\sigma_{\max}$ (MPa)
A	57.06	249.70
B	72.03	187.85
C	42.75	249.84
D	87.3	243.36

Table 6: *A-D optimal designs: thickness and peak stress.*

One can see all designs other than Case C did not converge close to the lower thickness boundary 41.29mm (see Table [6]). Results from Case A show the uniform plate thickness solution at $\sigma = 249.7\text{MPa}$ meaning the constraint is active. A smooth gradient can be seen in Case B (Figure [9]) due to the corner driven interpolation. Regardless, the case performed worse when compared to Case A counting an increase in mean thickness of 14.97mm . This shows a sharp spatial adaptation of the surrogate, unable to provide sensible smoothness. By locally reinforcing with such intensity, the design drifted away from the stress limit. Thus, significantly reducing peak stress to 187.85MPa at a cost of higher mass. In case C the mean thickness reaches the lowest optimum out of all cases, showing a 25% decrease (compared to the second lowest optimum). A mild smooth thickness variation at the boundary condition can be seen confirming a satisfactory generalisation of the surrogate model and optimiser (see Figure [10]). The smoothness of Case C aligns with realistic manufacturing and structural behaviour. Contrarily, Case D (Figure [11]) shows the sharpest spatial adaptation, where the optimiser thickened locally only, where the surrogate predicted hot spots. This shows a clear overdesign making Case D the least optimal configuration due to its average thickness of 87.3mm (2x Case C’s mass). The surrogate mimicks local reinforcement strategies used in structural engineering, however does not apply enough smoothness to the solution, resulting in excessively thick tiles. Cases B to D all reinforced the bottom left corner of the plate where boundary conditions are applied. Further, showing the surrogate model’s ability in prediction of stress concentration regions.

Finally this study shows increasing the resolution of the optimisation yields more spatial adaptation ($A < B < C < D$). In the scope of the task at hand, the optimum design from Case C is the most practical compromise offering the lightest weight with an honest spatial adaptation and high manufacturability. Unfortunately, the most promising design case (D) overfitted fine-scale stress, resulting in an excessively heavy plate design. This behaviour was unexpected considering the high fidelity the surrogate proved during previous tasks. Case D would benefit from introducing a more robust filtering of the surrogate stress gradients to avoid non-physical oscillatory stress patterns.

4: Digital Twinning

4.A Sensor Configuration

Early signs of damage in the beam can be detected most efficiently by coupling different results deriving from the beam’s strain, vibrations and acoustic emissions mainly. Intricate, yet high fidelity processes such as *Digital Image Correlation* are not suitable for aircraft integration. Conversely, implementing a simpler device like strain gauges would suit better the problem at hand. Placing strain gauge sensors near the fixed end of the cantilever plate (wing root) is the most practical and effective option. The placement site would be where maximum bending moments and *von-Mises* stresses occur due to combined loading; providing critical information with regards to the failure of the beam. Monitoring changes in strain indicates load, compression and tension experienced by the beam. Furthermore, distributing *Accelerometers* across the plate, particularly at the free end (wing tip) and mid span, would capture the wing’s beam’s dynamic responses. The data acquired is essential for modal analysis to identify stiffness degradation; allowing frequency and mode shape extraction(5). Most importantly, the vibration sensor allows engineers to identify early stiffness degradation. Lastly, an *Acoustic Emission Sensor* should be placed strategically near known high-

stress concentration regions. Thus, allowing the detection of micro-cracks and fracture propagation based on sound waves, enabling early damage diagnosis.

4.B AI Implementation

AI leverages real time sensor data for predictive maintenance and defect detection in the beam. Applying a *Bayesian FFT* method on acceleration data to perform *Operational Model Analysis* is optimal for vibration and dynamic monitoring. The model operates by mapping changes in identified modal parameters (frequency, damping ratio, modal shape) to a *Digital Twin* model to reveal degradation or stiffness loss in the beam (5). Additionally, applying a *Principal Component Analysis* to continuous strain and displacement data is essential for anomaly detection. Data fluctuation residuals are traced by a *Squared Prediction Error* indicator which, filters out environmental noise and flags sudden shape changes in strain or displacement. Further, detecting potential structural damage when these values exceed a defined threshold (5). Finally, a deep learning model (such as a 3D *CNN*) should be employed to analyse raw sensor data or time-sequential patterns for defect classification and/or damage prediction.

4.C Maintenance Decisions

A *Digital Twin* is characterised by its ability to be dynamically updated with sensor data to inform decisions that realise value (6). Real-time data (modal properties, strain and displacement) by the *Structural Health Monitoring* system is used to validate and calibrate the *Finite Element* based surrogate model of the plate. This continuous feedback loop ensures an accurate representation of the beam's recorded structural behaviour. Methods like *Bayesian Statistical Inference* can estimate structural states at unmonitored positions, enhancing the extensive support for maintenance. *AI* diagnostics and the surrogate model are used by the *Digital Twin* to provide predictive and prescriptive insights. Thus, allowing for *Condition-Based Maintenance*, a separate approach from archaic reactive repair procedures. Thus, optimising maintenance work-scoping and prediction of new parts needed. This process allows a reduction in downtime. Most importantly, mitigating catastrophic failures.

4.D Maintenance Decisions

Deploying *AI* on an aircraft structure requires addressing technical and operational problems associated to complex industrial systems (7). Given the structure's high data volume and safety-critical nature, an Edge *AI* is necessary. Diagnosing damage or adapting control must be made in microseconds near the data source rather than relying on cloud processing. Challenges include managing and updating *AI* models across numerous Edge devices. Furthermore, hardware constraints are an additional hurdle as they limit the model complexity. Finally, issues such as sensor drift and missing data are inherent in real-world environments and must be addressed to maintain the model's accuracy.

4.E Interpretability, Transparency, Ethics Safety

Ensuring trustworthiness is paramount for safety-critical decisions. Transparency demands that both the processes and outcomes are justifiable. This requires implementing Explainable *AI methods* to clarify the process of these "black-box" predictions (8). Further, requiring coherent and plain explanations from competent authorities. Furthermore, there are major issues with achieving a certification for these algorithms which progressively evolve. The complexity of these *AI* systems creates an accountability gap, making questions arise with regards to the *AI*'s responsibility in case a false positive is identified. Most importantly, embedding *AI* in structural components turns every structure in a potential, physical cyber-attack surface. Therefore, meaning the system must have robust defence lines to malicious data corruption or poisoning, which could lead to incorrect predictions.

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